DEEP REINFORCEMENT LEARNING FOR RAN

RANOpt AI: From RL Foundations to Production PPO

DQN, PPO, the RANOpt AI decision engine, and real-world RAN use cases — built for RF engineers

RANOpt AI — Product Vision

An interactive, AI-assisted 5G/6G RAN decision platform

The problem

RF engineers manually balance competing KPIs — throughput, coverage, fairness, energy — across cells and traffic conditions, largely by intuition and trial-and-error.

The approach

RANOpt AI is a browser-based platform that uses an AI Decision Engine to recommend configurations, making the tradeoff systematic, explainable, and fast.

Built by Trilcomm Technologies as one of two primary products, alongside the 5G NR simulator pipeline. Designed for RF engineers as intended users — decision support layered on top of engineering expertise, not a replacement for it.

From RL to DRL: Why It Matters to You

Reinforcement Learning (RL)

An agent learns by trial and error: observe the state, take an action, get a reward, adjust, repeat. The simplest version keeps a lookup table — one best action per state.

Deep RL (DRL)

DRL swaps that lookup table for a deep neural network — because the table approach collapses once the network is even moderately realistic.

A real cell has SINR, PRB utilization, throughput, latency, active UE count, neighbor interference, and time-of-day traffic. Multiply across a cluster of cells and the state space becomes astronomically large — far too big for a lookup table, and a neural network generalizes to states it has never exactly seen before.

DQN — Deep Q-Network

Best for: choosing among a fixed set of discrete options

How it thinks

DQN scores every possible action in the current state — a "Q-value" — and always picks the highest-scoring one. It needs the action menu defined up front.

Where it fits in RAN

Selecting a fixed antenna tilt preset • Picking a modulation/coding scheme from a table • Choosing a handover target from a candidate cell list

Strengths

- Simple to interpret — ranks discrete choices
- Fits naturally categorical decisions
- Easy to cross-check against known-good configs

Limits

Struggles with continuous parameters (e.g., a power level anywhere between 0–43 dBm) since every option must be enumerated ahead of time.

PPO — Proximal Policy Optimization

Best for: continuous or multi-parameter, multi-objective decisions

How it thinks

PPO learns a policy directly — network conditions in, action out — rather than scoring individual choices. It updates cautiously, in small trusted steps, which is why it trains more stably than older policy-gradient methods.

Where it fits in RAN

Fine-grained power control • Continuous tilt adjustment • Joint power + tilt + scheduling-weight tuning — matching how most real RAN problems actually look

In practice: RANOpt AI

Trilcomm's browser-based AI decision engine uses PPO as a single-step contextual bandit over a 5-dimensional 6G RAN context vector:

- Cell DL Throughput
- Cell Edge Throughput
- PRB Utilization
- Fairness (Jain's Index)
- Energy Efficiency

One explainable recommendation balancing all five KPIs at once — not a black-box multi-step control loop.

DQN vs. PPO — Choosing the Right Tool

DQN

Discrete choices

- Use when the decision is a pick from a fixed, enumerable menu
- e.g., one of several fixed tilt presets, one of several handover targets
- Simple, interpretable ranking of options

PPO

Continuous & multi-objective

- Use for continuous or multiple simultaneous parameters
- e.g., joint power + tilt + scheduling-weight tuning
- Weighs tradeoffs across a composite reward (throughput vs. fairness vs. energy)

In practice, PPO is the more widely used choice for RAN resource management and scheduling — most real engineering decisions aren't neatly discrete, and PPO's stable training makes it easier to deploy with confidence.

A Worked Example: Building the Reward Signal

How PPO turns multiple KPIs into one training signal

The setup

Consider an eMBB-only 5G cell (no URLLC) tracked on four KPIs: Throughput, SINR, PRB Utilization, and Latency.

Setting the weights

For an eMBB-only context, throughput, SINR, and PRB utilization dominate ($w_1 \approx w_2 \approx w_3$), while latency carries much less weight (w_4) since there's no strict low-latency service class to protect.

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reward = w₁·Throughput + w₂·SINR +
w₃·PRB_Utilization − w₄·Latency
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All KPIs combine simultaneously in one weighted sum, normalized to a common scale — not evaluated sequentially.

Cold-start pretraining: without historical KPI logs, a browser-based simulator can bootstrap PPO using physics-based RF equations — link budget, Shannon-Hartley capacity, PRB scheduling logic — before any live data exists.

RANOpt AI — Architecture at a Glance

AI Decision Engine (deployed)

Inference-only. Takes the live five-KPI context vector and returns one explainable configuration recommendation — no training happens at inference time.

PPO Training Demo (illustrative)

A separate, visible mode showing the PPO learning process itself — kept distinct from the deployed engine so a demo of "how training works" is never confused with production inference.

Formulated as a single-step contextual bandit, not a full multi-step MDP — no chained state transitions. This is a deliberate, corrected simplification: earlier drafts overclaimed a full MDP; the distinction was clarified in the INDICON 2026 paper submission.

Practical Use Cases in RAN Planning & Optimization

Antenna tilt & azimuth

Balancing cell-edge coverage against inter-cell interference across many cells and traffic conditions.

Power control

Adjusting transmit power per cell or user to manage interference as traffic and channel conditions shift.

Load balancing & handover

Deciding when and where to hand users between cells to avoid overloaded sectors during surges.

Energy efficiency

Scaling power or sleeping underused carriers in low-traffic periods without hurting peak performance.

Multi-KPI tradeoffs

Balancing throughput, fairness, coverage, and energy explicitly instead of by intuition alone.

Systematic exploration

Searching the tradeoff space at a scale and speed no manual process can match.

Where the Industry Is Headed

O-RAN RIC

O-RAN's RAN Intelligent Controller (RIC) architecture standardizes exactly this pattern: AI-assisted decision support layered onto the RAN via open interfaces, rather than a closed, vendor-specific black box.

6G AI-native vision

6G roadmaps treat AI/ML as a native network capability rather than an add-on — continuous learning and optimization built into the RAN itself.

RANOpt AI is built in this direction: an explainable, engineer-facing decision layer, not a replacement for RF engineering judgment.

Why This Matters — Not Just for Data Scientists

DRL doesn’t replace RF engineering judgment. It automates the systematic exploration of a tradeoff space engineers already understand intuitively from years of field experience.

Encode your judgment

The value is in the reward function — defining what "good" looks like across KPIs — then letting the algorithm search at scale.

Explainability matters

A good DRL decision engine shows which KPIs drove a recommendation, so engineers can validate it against experience.

RANOpt AI puts this into practice

A single-step contextual PPO engine over five RAN KPIs, with an INDICON 2026 paper formalizing the approach.